

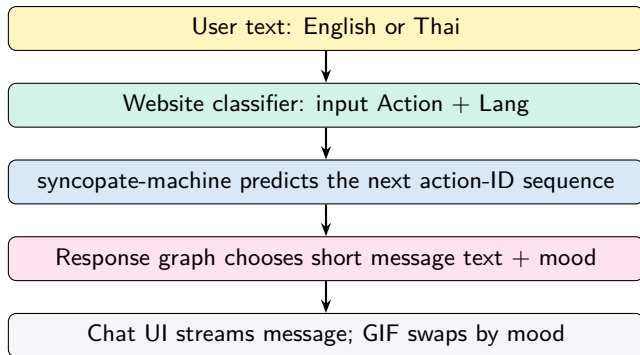
Syncopate-Machine

Tiny Softmax Transformer for Action-Sequence Chat

Dancing With My Code

Softmax rollback, 2026-05-26

The Contract



The model does not generate words. It predicts conversation actions.

Action-ID Flow

Input Side

- Website classifies user text
- Language is explicit: Thai or English
- Model input is compact integer IDs
- Context length stays at 64

Output Side

- Model predicts next action IDs
- Response graph writes final words
- Mood controls the mascot GIF
- Chat stays short and punchy

This is a tiny behavior brain for the website.

Action Vocabulary

ID	Action	Role	ID	Action	Role
0	Start	sequence start	12	GoRust	Rust beats Go
1	End	sequence end	13	RustWin	Rust wins again
2	Greeting	hello	14	GoBully	roast Go
3	ReplyGreeting	hello back	15	WhoMe	identity
4	Eat	food topic	16	AboutCreator	about Lookhin/site
5	DailyLife	normal chat	17	PissOff	clap back
6	AskBack	keep talking	18	CalmDown	de-escalate
7	Joke	quick joke	19	Farewell	goodbye only
8	Roast	playful insult	20	Normal	neutral mood
9	Happy	happy mood	21	TH	Thai
10	Question	question	22	EN	English
11	Insult	user swears			

Tiny Model Shape

Browser Model

Vocab	23
Context	64
Layers	1
Width	64
Attention heads	4
KV heads	1
FFN width	64
Params	24,192

Decoder Block

- RMSNorm
- Causal scaled dot-product softmax
- RoPE position rotation
- Grouped-query attention
- SwiGLU feed-forward
- Weight-tied output projection

The neural model decides **what kind of reply happens next**; the website decides **what words hit the chat box**.

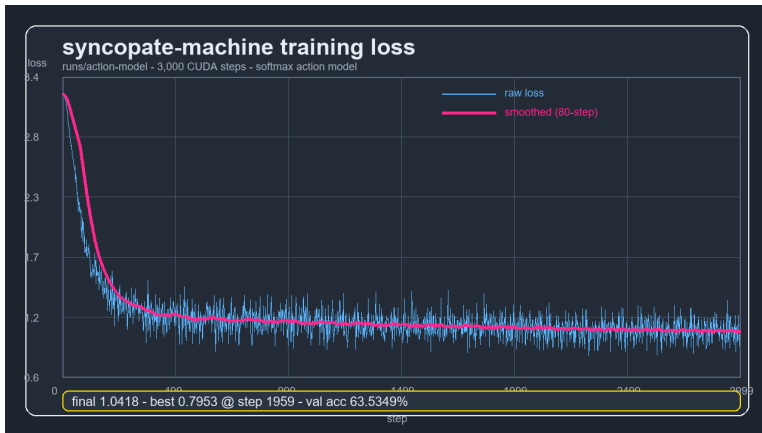
Softmax Attention

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right), \quad O = AV$$

- This is the only active attention path now.
- For this tiny model, softmax is faster/saner in practice.
- The checkpoint with the best measured validation accuracy is softmax.
- GQA keeps the browser checkpoint compact.

Boring math, useful model. Ship it.

Training Loss Curve



Raw CUDA training loss with 80-step smoothing from runs/action-model/loss.csv.

Parameter Count

Config: $V = 23$, $N = 1$, $d = 64$, $n_h = 4$, $n_{kv} = 1$, $d_k = 16$, $d_{ff} = 64$

Component	Formula	Params
Embedding	$V \times d = 23 \times 64$	1,472
Attention per layer	$d^2 + 2dn_{kv}d_k + d^2$	10,240
FFN per layer (SwiGLU)	$3dd_{ff}$	12,288
Norms per layer	$2d$	128
Final norm	d	64
Total	$1472 + 1(10240 + 12288 + 128) + 64$	24,192

Output projection reuses the embedding matrix, so it adds zero new params.

Current Checkpoint

Metric		Value
Checkpoint	runs/action-model/final.mpk	
Params		24,192
Validation loss		1.113533
Validation perplexity		3.0451
Validation accuracy		63.5349%

Website Asset

This checkpoint is copied to dancing-with-my-code-v2/assets/model-personal-v2.mpk.

Training Path

Data Build

```
python examples/build_action_data.py --offline
```

CUDA Training

```
cargo run --release --features cuda --example train_action_model --  
  --steps 3000 --batch-size 32 --lr 0.003  
  --checkpoint-dir runs/action-model
```

- Training is CUDA-only by design.
- Website runtime can choose WebGPU or CPU.
- Training data is compact action-ID sequences.

Browser Runtime

- Website loads model-personal-v2.mpk
- Website loads model-config-personal-v2.json
- Runtime tries WebGPU first when requested
- Auto mode falls back to CPU when WebGPU is unavailable
- Chrome without a valid WebGPU adapter should not kill the chat box
- UI streams message text from the response graph

Mood Contract

Model output action sequence maps to one mood: normal, happy, frustrated, or sad. The mascot GIF changes from that mood.

Deploy Shape

- Build the Leptos site with `trunk build --release`
- Ship the generated `dist/` directory to Vercel
- Keep model assets in `dancing-with-my-code-v2/assets/`
- Cache `.mpk` and `.json` assets aggressively

Final State

Softmax drives the tiny action model. The website owns the voice.

References

- Burn framework: <https://github.com/tracel-ai/burn>
- RoPE / RoFormer: rotary position embeddings
- GQA: grouped-query attention
- SwiGLU: gated feed-forward block
- Project docs: README.md and DEVLOG.md